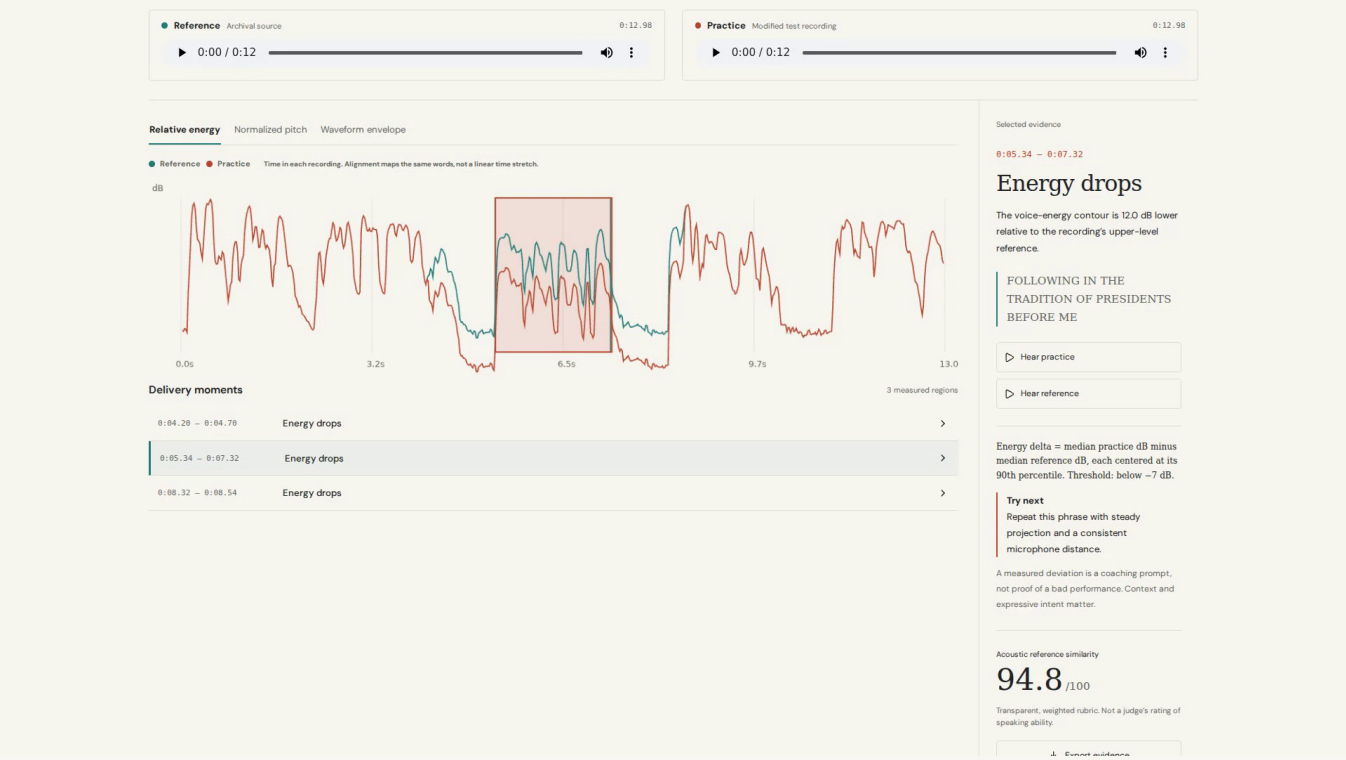
A moment worth practicing

Working local inference. Exact intervals. Explainable feedback.

Student speakers and coaches receive broad advice such as “improve your pacing” without knowing which phrase to rehearse. CadenceLab compares recordings of the same English words and turns acoustic differences into timed evidence. Select a region, play either recording, inspect its numerical explanation and export JSON.



The score is reference similarity, not talent, truth, emotion or rhetorical quality. A reference is a comparison target, not a universal expressive ideal. Every recording keeps its own timeline; matched words provide correspondence.

The shipped interface includes controlled examples, a real audio/transcript upload flow, a corpus explorer and inclusive evaluation. No backend account or audio-upload endpoint is required.

A contrastive, labeled corpus

90 paired cases · 5 excerpts · 2 speakers · 3 archival speeches

Four Obama excerpts come from official January 7 and January 14, 2017 weekly addresses; a JFK excerpt comes from the 1961 inaugural speech. The actual JFK bytes use the pinned Xenova mirror. Sources and rights are documented; no affiliation or endorsement is implied. Unverified FDR candidates were excluded.

Modification	Subtle	Moderate	Strong
Local pace	1.15×	1.40×	1.80×
Local gain	0.50	0.25	0.10
Added pause	0.4 s	0.9 s	1.8 s
Noise SNR	20 dB	10 dB	0 dB
Local pitch	+2 st	+4 st	+7 st

Each excerpt adds three controls: unchanged, whole-recording gain and whole-recording pitch. Five excerpts × eighteen variants = ninety cases. Exact sample-level injection boundaries are ground truth, independent of detector predictions. FLAC hashes, durations and alignment bounds are checked.

The full official Obama transcripts are aligned before excerpt cropping. Audio is serialized and read back before analysis. FFmpeg atempo and rubberband perform tempo/pitch changes; gain multiplies samples, pauses insert zeros and Gaussian noise uses a fixed seed.

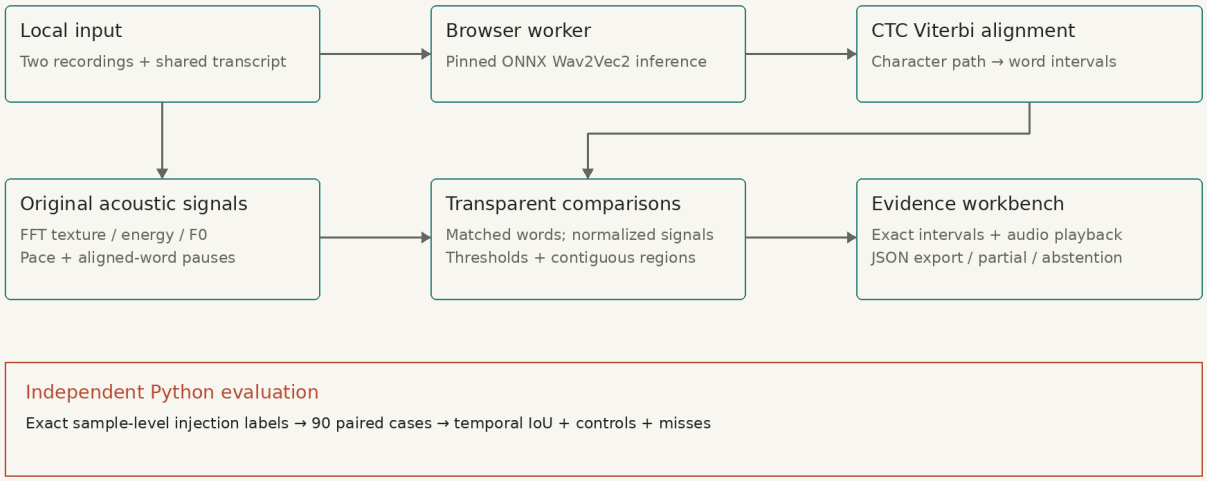
Limit: synthetic historical English male speakers do not establish real-student accuracy or fairness across accents, genders and recording devices. Thresholds were developed with this corpus.

The inference and evidence path

A real acoustic model followed by inspectable numerical rules

CadenceLab architecture

Original audio stays on-device. Evidence and uncertainty remain inspectable.



No audio-upload endpoint. Model hosts receive ordinary asset requests on first use.

Pinned Wav2Vec2 quantized ONNX weights (95,286,046 bytes) supply frame-level probabilities. Transcript-constrained CTC Viterbi alignment includes blank states and repeated-character constraints. Four-second windows have 0.5-second context; central emissions discard overlap. Frame centers use the model convolution stride and receptive field.

Automatic gain conditioning affects alignment input only. Original waveform samples remain untouched for acoustic evidence. Python and the browser worker share the alignment and normalization approach; browser ONNX inference runs via WebAssembly.

The first analysis downloads model/runtime assets. Uploaded recordings and transcript remain local. The model host observes ordinary asset requests. Browser extensions or compromised devices sit outside this privacy boundary.

A transparent reference rubric

Measured differences become regions with reasons and actions

Evidence	Threshold	Weight
Matched-word pace	$\lvert\log_2 \text{ratio}\rvert > 0.35$	30%
Relative energy	Below −7 dB	25%
Spectral flatness	Difference > 0.05	15%
Normalized pitch	Difference > 4 st	15%
Aligned pause gap	Added gap > 0.9 s	15%

40 ms windows and 20 ms hops support energy, FFT texture and autocorrelation F0. Python additionally computes thirteen MFCCs. Pitch is median-centered in semitones with a robust matched-word offset; energy is centered at each recording’s 90th percentile. Local pace compares corresponding nearby word windows.

Adjacent flagged words form contiguous regions. The similarity score subtracts the weighted proportion of time above thresholds. It is a documented rule-based rubric, not a learned human grade.

When transcript support becomes unreliable, a frame-level noise detector compares practice spectral flatness with the reference speech-frame 95th-percentile baseline + 0.05. Such partial reports withhold word-based scores and do not assert precise word correspondence. Other unsupported cases abstain explicitly.

Evaluation, including the misses

Declared matching threshold: union temporal IoU ≥ 0.30

Diagnostic	Result
Altered cases localized, all severities	43 / 75
Moderate + strong cases localized	40 / 50
Mean temporal IoU, including misses	0.403
Controls with false regions	2 / 15
Unchanged + global-gain controls with regions	0 / 10
Complete abstentions / partial reports	4 / 2

Inclusive results retain subtle misses, false positives, abstentions and partial reports. Two false detections occur in whole-recording pitch controls, so full speaker independence is not established. Stronger noise is detectable through frame texture even when phonetic alignment fails.

These are development-set diagnostics, not a held-out estimate. Detector thresholds were developed with these examples. Moderate 0.9-second pauses sit near the 0.9-second rubric threshold; subtle changes often remain below thresholds. There is no consent-based real-student study yet.

Full case-level output, transformation labels, source hashes and group-level diagnostics are public in the repository. A future independent test set and human coaching comparisons are required before claims of general performance.

Verification and reproduction

Public source, clear setup and a disclosed production workflow

Five pipeline tests pass: repeated-character CTC behavior and precise frame centers; normalization; unchanged evidence; transcript-match requirements; and all 90 audio hashes, durations and interval bounds. A genuine browser upload recovered three energy regions. Its score matched Python at 94.8, with maximum boundary differences of 20 ms.

Automated accessibility checks found no axe violations across Compare, Dataset, Method and Upload. Mobile overflow and page-error checks pass. Dependency audit reports zero known vulnerabilities at the recorded check. Input constraints, escaped text, security headers and local-only network behavior are documented. This is not a security certification.

README: install pinned Python dependencies, download public inputs, run pipeline tests, install web dependencies and build with Vite. The public corpus supports immediate labeled-example testing. The project includes model provenance, rights, security boundaries, AI assistance disclosure and HyperFrames video code.

Demo: 303 seconds at 1920×1080 and 30 fps, with a real app recording, local upload inference, synchronized captions and a same-volume audio comparison. Stock synthetic narration is disclosed. No paid video generator was used.

Primary references:
Official event rules: multimodal-ai-hackathon-2026-7.devpost.com/rules
Model: huggingface.co/Xenova/wav2vec2-base-960h
JFK transcript: archives.gov/milestone-documents/president-john-f-kennedys-inaugural-address
Obama sources: obamawhitehouse.archives.gov (2017/01/07 and 2017/01/14)
Source: github.com/adeelfalaksher900-creator/cadencelab-mah2026